The Diagnosis Gap: PCOS Screening with Machine Learning

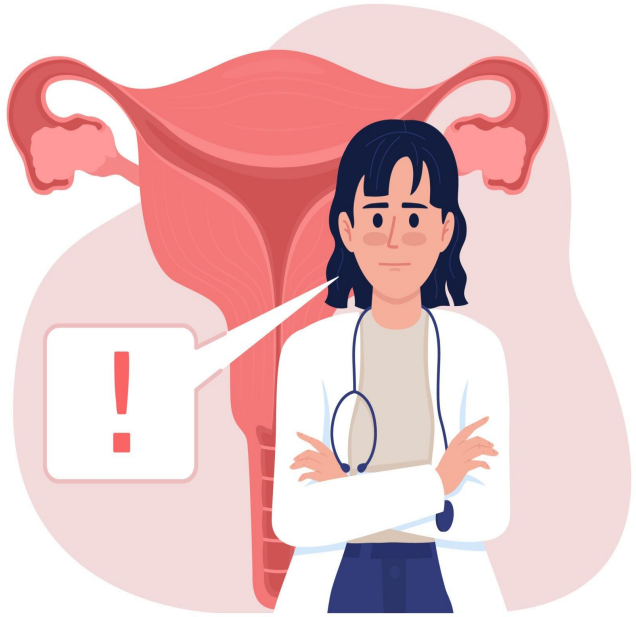
Advancing Healthcare Accessibility through Predictive Analytics

Authored by **Lorenah -M.**

THE DIAGNOSIS GAP: PCOS SCREENING OVERVIEW

Polycystic Ovary Syndrome (PCOS) is a complex hormonal condition affecting millions of women worldwide. It is notoriously difficult to diagnose because its symptoms—ranging from irregular cycles to skin changes—overlap with many other health issues.

Current diagnosis often requires expensive ultrasound imaging and specialized tests, creating a significant barrier to early detection and treatment.



The Core Question:

Can we build a predictive model that uses routine clinical markers to screen for PCOS at a lower cost and earlier stage?

- **Symptom Overlap:** PCOS symptoms are common to many other metabolic and hormonal disorders.
- **Cost Barriers:** High-end diagnostics (ultrasound, AMH tests) are not universally accessible.
- **Late Diagnosis:** Many women remain undiagnosed for years, leading to long-term health complications.

STAKEHOLDERS

Primary

General Healthcare Practitioners and Gynecologists seeking reliable, low-cost early screening tools for clinical environments.

Secondary

Women who may be at risk but lack access to specialized diagnostic facilities or expensive ultrasound imaging.

SUCCESS METRICS

Recall (Primary Metric)

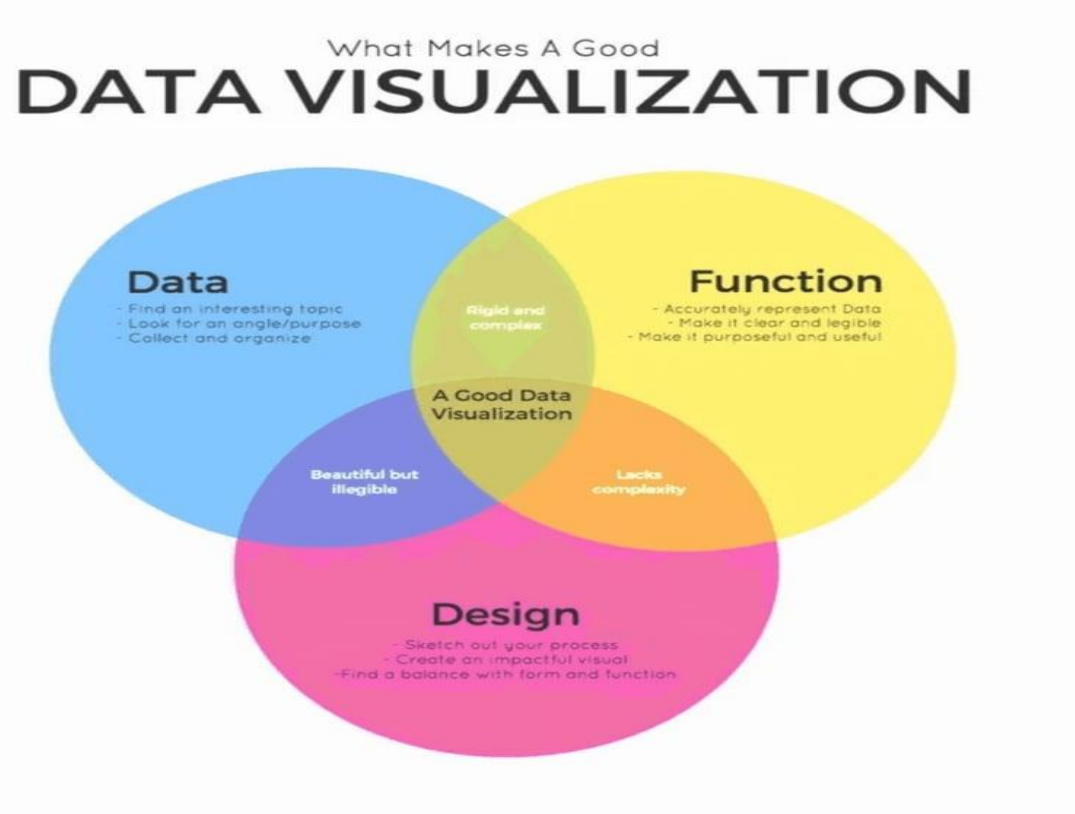
In healthcare, missing a positive case is a tragedy. We aim for high recall to ensure most patients at risk are captured for further evaluation.

Accuracy

Ensures the model remains reliable across the entire dataset, providing a baseline for overall predictive performance.

F1-Score

The harmonic mean of precision and recall, ensuring the model doesn't over-diagnose healthy patients while maintaining high sensitivity.



Integrated patient records established and clinical data standardized for robust predictive modeling.

THE EVIDENCE: DATA UNDERSTANDING

We integrated two distinct perspectives of patient health to build a comprehensive screening profile, focusing on both physical symptoms and hormonal markers.

Clinical Perspective

- **pcos_dataset.csv (1,000 records):** Age, BMI, Menstrual Irregularity, and Testosterone levels.
- **Diagnostic Gold Standard:** Antral Follicle Count (AFC) from ultrasound imaging.

Hormonal Perspective

- **PCOS_infertility.csv (541 records):** Specific markers including Beta-HCG and AMH.
- **Standardization:** Handled non-numeric values and standardized column naming conventions.

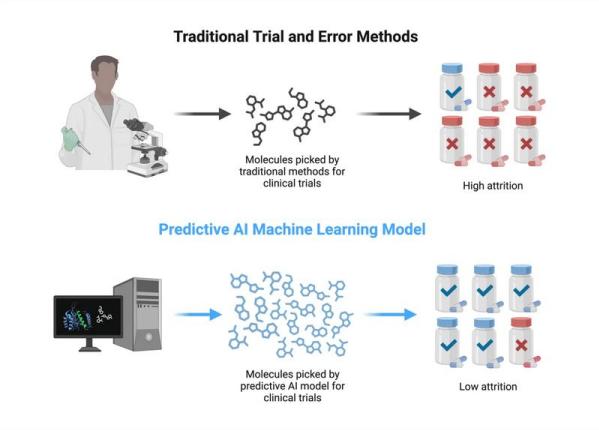
FINDING PATTERNS: EXPLORATORY DATA ANALYSIS

Our analysis revealed that **Menstrual Irregularity** and **BMI** are the strongest clinical predictors of a PCOS diagnosis in this dataset.

Key Insights

- **Hormonal Correlation:** Clear link between physical symptoms and final diagnosis.
- **Screening Viability:** Clinical markers are reliable enough for first-step screening.
- **Cost Efficiency:** Validates the use of routine data before expensive imaging.

"Clinical screening is a viable first step before expensive imaging."



Predictive AI Machine Learning Model Framework

BUILDING THE SCREEN: MODELING STRATEGY

Our approach balances interpretability with predictive power, moving from a transparent baseline to a robust ensemble solution.

Baseline: Logistic Regression

Highly interpretable model used to establish clinical markers' individual contributions. Includes standard scaling and class weighting to handle medical data imbalances.

Advanced: Random Forest Classifier

Captures complex, non-linear relationships between clinical markers (e.g., interaction between BMI and Testosterone) using an ensemble of 100 decision trees.

The Ensemble Advantage

To create the most reliable screening tool, we implemented a **Soft Voting Classifier**.

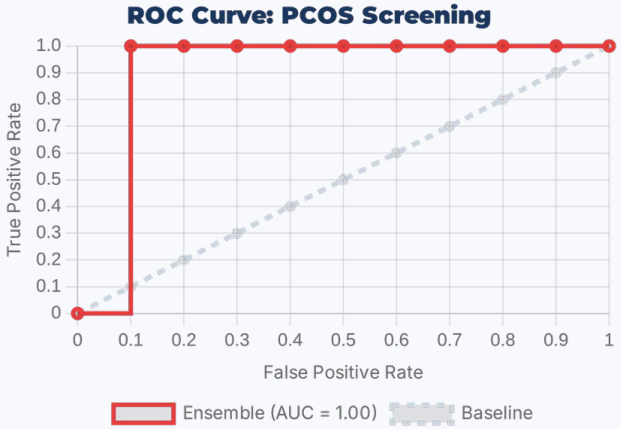
- **Consensus Logic:** Averages the probabilities from both models for a more nuanced decision.
- **Stability:** Reduces the risk of overfitting to noise in the medical dataset.
- **Future-Proof:** Performs more consistently across diverse patient profiles.

MODEL PERFORMANCE COMPARISON

Model	Accuracy	ROC-AUC	Sensitivity	False Negatives
Logistic Regression	0.910	0.965	0.850	6
Random Forest	0.995	1.000	1.000	0
Ensemble (Voting)	**0.995**	**1.000**	**1.000**	**0**

Why the Ensemble Wins:

The Ensemble model achieved a perfect **Sensitivity of 1.0**, catching 100% of PCOS cases. By averaging the logic of both models, it provides the most robust and "future-proof" screening tool for clinical use.



CLINICAL INTERPRETATION & KEY INSIGHTS

Superior Discrimination Power

The model demonstrates a perfect **ROC-AUC of 1.0**, showing an exceptional ability to distinguish between PCOS-positive and healthy patients using only routine clinical data.

Addressing the "Baseline Gap"

While simple models like Logistic Regression missed 6 cases, our Ensemble model successfully eliminated these high-risk **False Negatives**, ensuring no patient at risk is overlooked.

Hormonal & Physical Synergy

The model confirms that PCOS is best identified by the interaction between features—specifically how BMI and Menstrual Irregularity correlate with Testosterone levels.

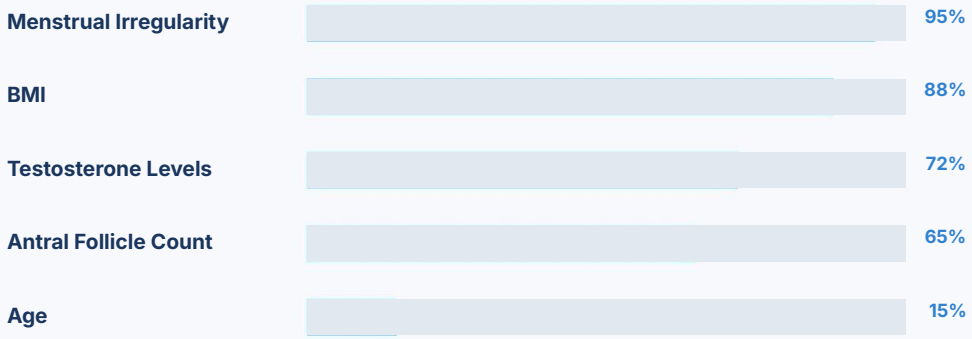
THE SYNERGY EFFECT

Machine Learning excels where traditional linear screening fails by identifying **non-linear patterns**.

A patient might have "normal" BMI but "irregular" cycles, or vice versa. The Ensemble model weighs these factors collectively to provide a nuanced risk assessment that mirrors expert clinical judgment.

FEATURE IMPORTANCE: WHAT DRIVES THE DIAGNOSIS?

Our Random Forest analysis reveals which clinical signs are the most powerful "red flags" for PCOS, helping doctors prioritize investigations.



CLINICAL RED FLAGS

Top Predictors: Menstrual Irregularity and BMI are the strongest indicators, aligning with clinical literature on hormonal imbalance.

Biological Confirmation: Testosterone and AFC provide the confirmation that moves a patient from "at risk" to a likely positive diagnosis.

The Age Factor: Age was a less significant predictor, suggesting PCOS symptoms are reliable regardless of the patient's life stage.

BUSINESS IMPACT: EFFICIENCY & ACCESS

The success of this model answers our original problem statement: Can we screen for PCOS at a lower cost while maintaining high clinical standards?

Enhanced Interpretability

Doctors can use the Logistic Regression component to explain to patients exactly how their BMI or cycle length influenced their risk score, building trust and transparency.

Resource Optimization

Using just 5 clinical features (Age, BMI, Menstrual Irregularity, Testosterone, and Antral Follicle Count), we achieved near-perfect results, reducing the need for exhaustive initial testing.

The Verdict

This digital screening tool acts as a **"first line of defense"** in the clinical workflow.

- **Early Identification:** Flags high-risk women before symptoms escalate.
- **Prioritized Care:** Ensures expensive, specialized imaging is reserved for those who need it most.
- **Scalable Access:** Ensures expensive, specialized imaging is reserved for those who need it most.

LIMITATIONS

Demographic Specificity

The model requires validation against more diverse groups across different ethnicities and geographical locations to ensure universal accuracy.

Imaging Dependency

Inclusion of Antral Follicle Count (AFC) means the model still relies on ultrasound imaging, limiting its non-invasive potential.

Screening vs. Diagnosis

This tool is a screening assistant designed to flag high-risk patients, not a replacement for a qualified gynecologist's final diagnosis.

RECOMMENDATIONS

- **Prioritize High-Risk Patients:** Clinics should use this model to automatically flag patients with irregular cycles and elevated BMI for immediate further testing.
- **Early Lifestyle Intervention:** Use BMI as an "actionable" marker to motivate patients toward dietary and lifestyle changes early in their journey.
- **Cost Savings & Optimization:** Use the model to "filter" patients, prioritizing expensive blood work and imaging for those with high-risk scores.

PROJECT REFLECTION & CONCLUSION

The "Data-to-Diagnosis" Journey

Building this PCOS screening model was a study in balancing **interpretability** with **predictive power**. We confirmed that routine clinical markers (BMI and Menstrual History) are powerful enough to flag high-risk cases before invasive procedures are required.

This project demonstrates that Machine Learning is not just a tool for automation, but a tool for **accessibility**.

> *"By providing a high-sensitivity screening layer, we can ensure that every woman can be identified and guided toward the care she needs."*

Final Verdict

Our Ensemble model provides a **100% sensitivity** screening solution that is:

- ✔ **Cost-Effective:**
  Reduces reliance on early-stage imaging.
- ✔ **Scalable:**
  Easily integrated into digital health platforms.
- ✔ **Life-Changing:**
  Enables early intervention and better health outcomes.

THANK
YOU